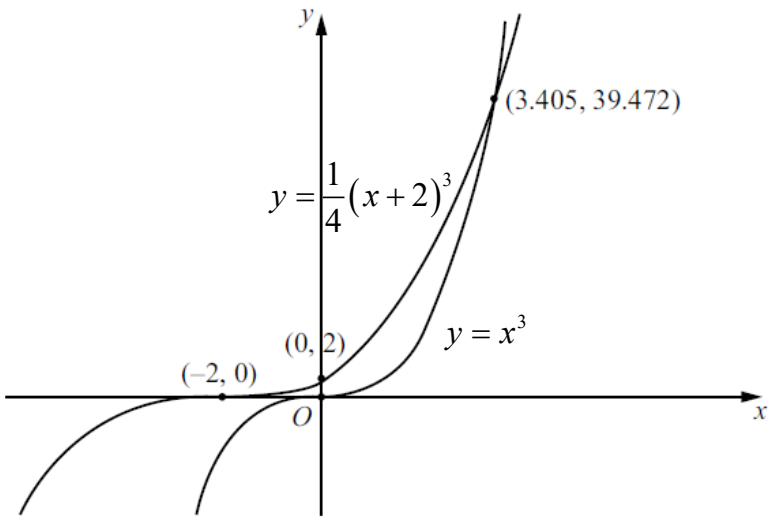
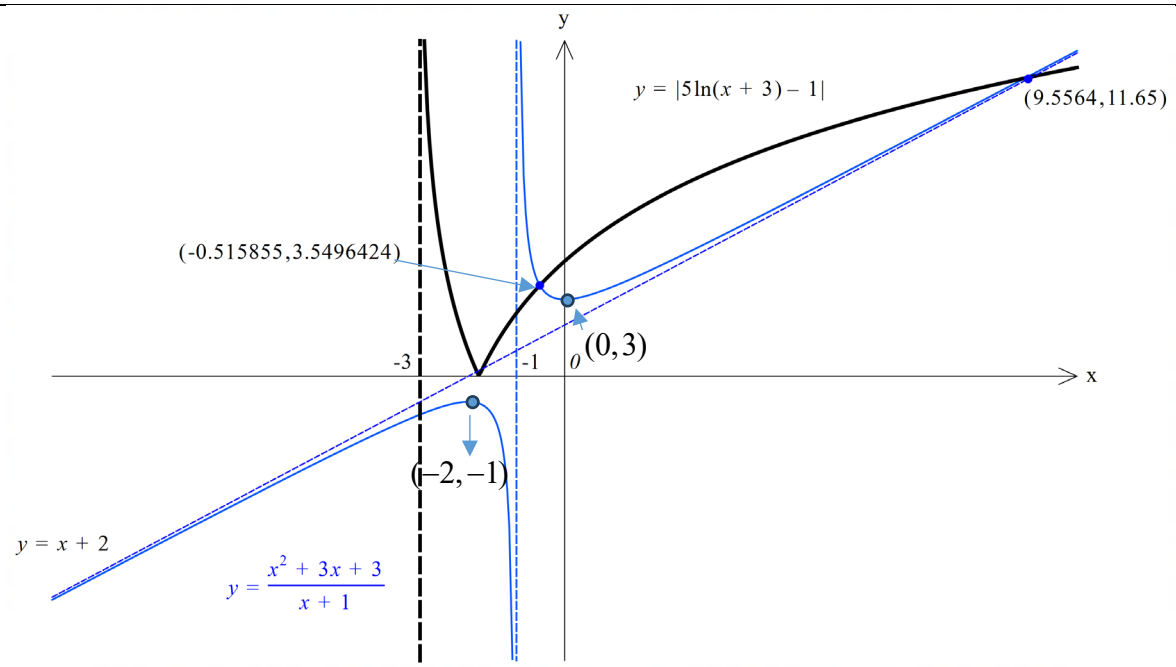
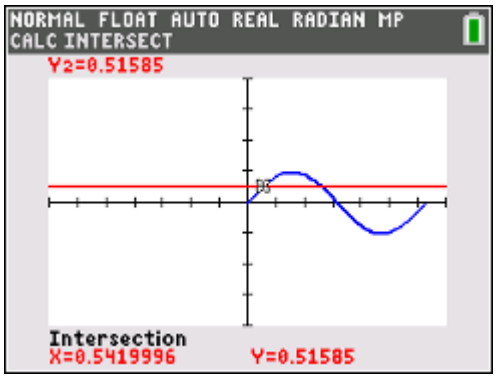
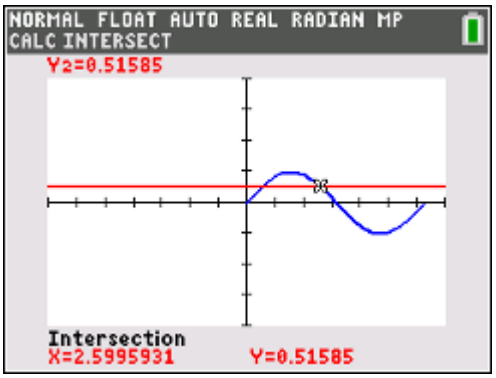
RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

H2 Math Year 6 Mock Paper 1: Solutions with Comments

No.	Suggested Solution	Remarks for Student
1	Let the circumference of the circle be C cm and the area of the circle be A cm^2. $C = \pi D$ $\frac{\mathrm{d}C}{\mathrm{d}t} = (\pi)\frac{\mathrm{d}D}{\mathrm{d}t}$ $= 0.3\pi$ $= 0.94248$ $= 0.942$ cm/s (correct to 3 s.f.) $A = \pi\left(\frac{D}{2}\right)^2 = \frac{\pi D^2}{4}$ $\frac{\mathrm{d}A}{\mathrm{d}D} = \frac{\pi D}{2}$ $\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{\mathrm{d}A}{\mathrm{d}D} \times \frac{\mathrm{d}D}{\mathrm{d}t}$ $= \left(\frac{\pi D}{2}\right)\left(\frac{\mathrm{d}D}{\mathrm{d}t}\right)$ When $r = 2$, $D = 4$ $\frac{\mathrm{d}A}{\mathrm{d}t} = \left(\frac{4\pi}{2}\right)(0.3)$ $= 1.88$ cm^2/s (correct to 3 s.f.)	Standard question requiring students to recall the relationship between diameter and area (circumference). The value of $\frac{\mathrm{d}C}{\mathrm{d}t}$ holds for all values of D (independent of D). Use of chain rule to find $\frac{\mathrm{d}A}{\mathrm{d}t}$

No.	Suggested Solution	Remarks for Student
2(a)	$y = x^3$ $\downarrow$ replace x by $x+2$ $y = (x+2)^3$ $\downarrow$ replace y by $4y$ $y = \frac{1}{4}(x+2)^3$	A translation of 2 units in the negative x-direction Followed by a scaling of factor $\frac{1}{4}$ parallel to the y-axis
2(b)	 x-coordinate of the point of intersection is 3.405 (3 d.p.)	Sufficient space and an appropriate scale needs to be chosen in order for the intersection point (3.405,39.472) to be displayed in the sketch. Note: The intersection point cannot be seen in the standard window of the GC.

No.	Suggested Solution
3(a)	 $y = \frac{x^2+3x+3}{x+1} = x+2+\frac{1}{x+1}$ Hence the graph has an oblique asymptote $y = x + 2$ and a vertical asymptote at $x = -1$.

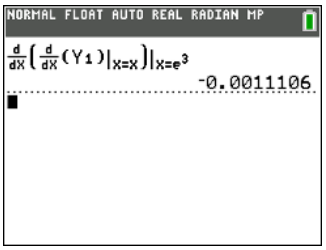
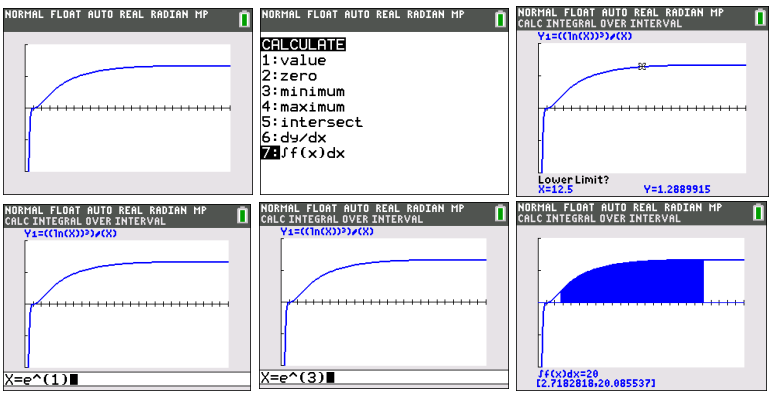
	The coordinates of the stationary points are $(-2,-1)$ and $(0,3)$.
(b)	We need to add the graph of $y=\lvert 5\ln(x+3)-1\rvert$ (which has a vertical asymptote at $x=-3$). The intersection points are $(-0.515855,3.5496)$ and $(9.5564,11.65)$. From the graph, the solution of $\dfrac{x^2+3x+3}{x+1}\geq\lvert 5\ln(x+3)-1\rvert$ is $-1<x\leq -0.516$ (3 s.f.) or $x\geq 9.56$ (3 s.f.)
(c)	Replacing x with $-\sin x$, we get $-1<-\sin x\leq -0.51585$ or $-\sin x\geq 9.5564$ $0.51585\leq \sin x<1$ or $\sin x\leq -9.56$ (N.A. since $-1\leq \sin x\leq 1$) $0.542\leq x\leq 2.60$ and $x\neq\dfrac{\pi}{2}$ (to exclude $y=1$) Note: Use of GC to sketch the graph of $y=\sin x$ to identify the part of graph that has $0.51585\leq y<1$  

No.	Suggested Solution	Remarks for Student
4(a)	$U_4=p+qU_3\Rightarrow 76=p+88q---(1)$ $U_5=p+qU_4\Rightarrow 70=p+76q---(2)$ Solving equation (1) and (2): $p=32, q=0.5$	
(b)	If the concentration converges to L, then taking the limit $t\to\infty$ in the relationship $U_{t+1}=32+0.5U_t$ we conclude that $L=32+0.5L$ and thus $L=64$.	As $t\to\infty$, $U_t\to L$ and $U_{t+1}\to L$ The linear equation is $L=32+0.5L$.
(c)	The concentration decreases and converges to 64.	When describing the behaviour of the sequence, we should describe if it decreases or

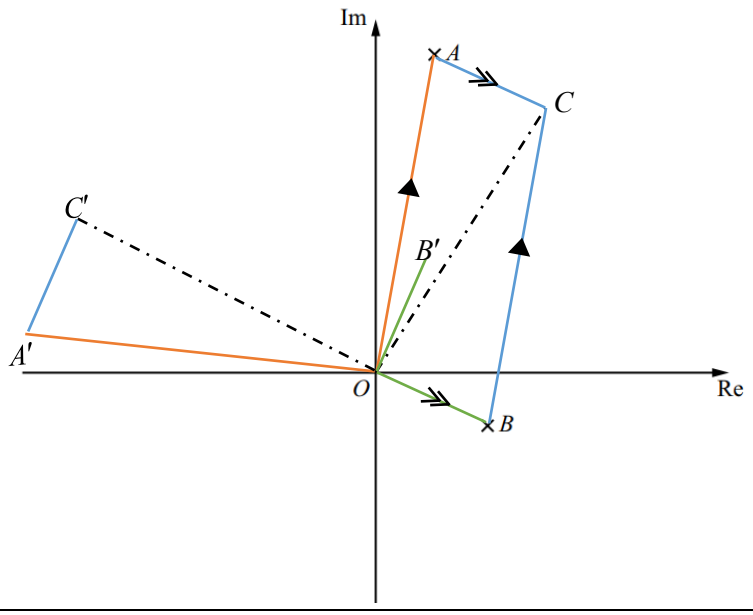
		increases, as well as whether it converges (to a limit which you should specify) or diverges. Using 'tends to', 'approaches' or the use of an arrow will not get full credit.

No.	Suggested Solution	Remarks for Student
5(a)	$y = \dfrac{(\ln x)^3}{x}$ $\dfrac{dy}{dx} = \dfrac{x\left[\dfrac{3(\ln x)^2}{x}\right] - (\ln x)^3[1]}{x^2} = \dfrac{3(\ln x)^2 - (\ln x)^3}{x^2}$ At turning point, $\dfrac{dy}{dx} = 0$ $\Rightarrow 3(\ln x)^2 - (\ln x)^3 = 0$ $\Rightarrow (\ln x)^2(3 - \ln x) = 0$ $\Rightarrow \ln x = 0$ or $\ln x = 3$ $\Rightarrow x = 1$ (rej $\because x > 1$) or $x = e^3$ Using the **1st Derivative Test** $\dfrac{dy}{dx} = \left(\dfrac{\ln x}{x}\right)^2 (3 - \ln x)$ Since $\left(\dfrac{\ln x}{x}\right)^2 > 0$ and $\ln x$ is increasing for $x > 1$, When $x < e^3 \Rightarrow \ln x < 3 \Rightarrow 3 - \ln x > 0$ When $x > e^3 \Rightarrow \ln x > 3 \Rightarrow 3 - \ln x < 0$ Therefore it is a maximum point at $x = e^3$.	The first derivative can be obtained using quotient rule or product rule for $y = \dfrac{1}{x}(\ln x)^3$. Students are reminded to reject $x = 1$, given that $x > 1$. Take note of the presentation of the first derivative test, which requires factorization of expression for $\dfrac{dy}{dx}$, followed by a discussion on the sign of the factor $(3 - \ln x)$.

x	$(e^3)^-$	e^3	$(e^3)^+$
$(3 - \ln x)$	+ve	0	−ve
$\dfrac{dy}{dx}$	+ve	0	−ve

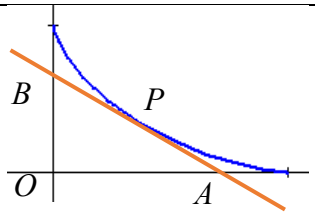
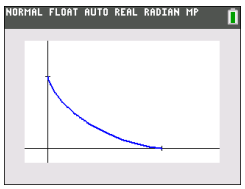
	OR Using the **2nd Derivative Test** $x^2\dfrac{dy}{dx}=3(\ln x)^2-(\ln x)^3$ Differentiate wrt x on both sides, $2x\dfrac{dy}{dx}+x^2\dfrac{d^2y}{dx^2}=\dfrac{6\ln x}{x}-\dfrac{3(\ln x)^2}{x}$ Since $x=e^3$ and $\dfrac{dy}{dx}=0$, $e^6\dfrac{d^2y}{dx^2}=\dfrac{6(3)}{e^3}-\dfrac{3(3)^2}{e^3}$ $\Rightarrow\dfrac{d^2y}{dx^2}=-9e^{-9}<0$ Therefore it is a maximum point at $x=e^3$.	Using the Second derivative test by applying product/quotient rule might look complicated to some students. A way around it is to use implicit differentiation. Otherwise, we can use the GC. 
(b)	$\text{Area}=\int_e^{e^3}\dfrac{1}{x}(\ln x)^3\,dx$ $=\left[\dfrac{(\ln x)^4}{4}\right]_e^{e^3}$ $=\dfrac{3^4-1}{4}$ $=20$ Note: Students should use GC to identify the required area. After obtaining the graph, students can use GC to check their answer.  Area = 20	Students should be able to recognize the form of the integral $\int f'(x)(f(x))^n\,dx$ $=\dfrac{(f(x))^{n+1}}{n+1}+C$ For those who are unable to, a last resort is to perform the substitution $y=\ln x$. Then $\int_e^{e^3}\dfrac{1}{x}(\ln x)^3\,dx$ $=\int_1^3(y)^3\,dy$ $[dy=\dfrac{1}{x}\,dx]$

No.	Suggested Solution	Remarks for Student
6(a)	$\mathbf{a}\times\mathbf{b}=\mathbf{c}\times\mathbf{a}$ $\mathbf{a}\times\mathbf{b}-\mathbf{c}\times\mathbf{a}=\mathbf{0}$ $\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}=\mathbf{0}$ $\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{0}$ Given that $\mathbf{a}\neq\mathbf{0}$ and $\mathbf{b}\neq-\mathbf{c}$, $\mathbf{a}$ is parallel to $(\mathbf{b}+\mathbf{c})$ $\therefore \mathbf{a}=k(\mathbf{b}+\mathbf{c}),\ k\neq 0$.	Properties of cross product are used in this question. Students are reminded that the RHS involves the zero vector **0**, to be written as $\underset{\sim}{0}$. The use of the conditions $\mathbf{a}\neq\mathbf{0}$ and $\mathbf{b}\neq-\mathbf{c}$ is important to guide students to the required solution.
6(b)(i)	Let $\mathbf{v}=\begin{pmatrix}x\\y\\z\end{pmatrix}$ $\mathbf{v}\times(\mathbf{i}-3\mathbf{k})=2\mathbf{j}$ $\begin{pmatrix}x\\y\\z\end{pmatrix}\times\begin{pmatrix}1\\0\\-3\end{pmatrix}=\begin{pmatrix}0\\2\\0\end{pmatrix}\Rightarrow\begin{pmatrix}-3y\\3x+z\\-y\end{pmatrix}=\begin{pmatrix}0\\2\\0\end{pmatrix}\Rightarrow\begin{cases}y=0\\z=2-3x\end{cases}$ $\mathbf{v}=\begin{pmatrix}0\\0\\2\end{pmatrix}+t\begin{pmatrix}1\\0\\-3\end{pmatrix},\ t\in\mathbb{R}$ The set of vectors $\mathbf{v}$ is $\left\{\begin{pmatrix}0\\0\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\0\\-3\end{pmatrix}\middle\|\ \lambda\in\mathbb{R}\right\}$.	Let $x=t$, $\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}t\\0\\2-3t\end{pmatrix}$ Students are reminded to learn how to express a collection of vectors using set notation.
(b)(ii)	The set of vectors are position vectors of points that form a line passing through the **point with position vector** $\begin{pmatrix}0\\0\\2\end{pmatrix}$ and is parallel to the vector $\begin{pmatrix}1\\0\\-3\end{pmatrix}$.	In describing a line geometrically, students are required to mention position vector of a point on the line and a direction vector for the line. Note: $\begin{pmatrix}0\\0\\2\end{pmatrix}$ is NOT a point.

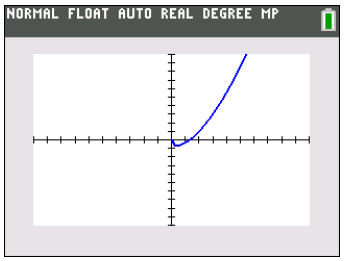
No.	Suggested Solution	Remarks for Student
7(a)	$2\mathrm{i}z+(1-2\mathrm{i})w=4 \quad \cdots\cdots(1)$ $(1+\mathrm{i})z+(2+\mathrm{i})w=3 \quad \cdots\cdots(2)$ $(1)\times(1+\mathrm{i}) \quad (1+\mathrm{i})2\mathrm{i}z+(1+\mathrm{i})(1-2\mathrm{i})w=4(1+\mathrm{i}) \quad \cdots\cdots(3)$ $(2)\times 2\mathrm{i} \quad 2\mathrm{i}(1+\mathrm{i})z+2\mathrm{i}(2+\mathrm{i})w=3(2\mathrm{i}) \quad \cdots\cdots(4)$ $(3)-(4) \quad (1+\mathrm{i}-2\mathrm{i}+2)w-(4\mathrm{i}-2)w=4+4\mathrm{i}-6\mathrm{i}$ $(3-\mathrm{i}+2-4\mathrm{i})w=4-2\mathrm{i}$ $w=\dfrac{4-2\mathrm{i}}{5-5\mathrm{i}}$ $=\dfrac{2}{5}\left(\dfrac{2-\mathrm{i}}{1-\mathrm{i}}\right)\times\dfrac{1+\mathrm{i}}{1+\mathrm{i}}$ $=\dfrac{2}{5}\left(\dfrac{2-\mathrm{i}+2\mathrm{i}+1}{1+1}\right)$ $=\dfrac{3+\mathrm{i}}{5}$ $z=\dfrac{4}{2\mathrm{i}}-\dfrac{(1-2\mathrm{i})}{2\mathrm{i}}\left(\dfrac{3+\mathrm{i}}{5}\right)$ $=-2\mathrm{i}-\dfrac{(-\mathrm{i}-2)(3+\mathrm{i})}{10}$ $=-2\mathrm{i}-\dfrac{(-3\mathrm{i}-6+1-2\mathrm{i})}{10}$ $=-2\mathrm{i}-\dfrac{-5-5\mathrm{i}}{10}$ $=\dfrac{1-3\mathrm{i}}{2}$	Students should attempt to eliminate either z or w. Since GC cannot be used in the question, detailed working for the multiplication and division of complex numbers are to be shown clearly, which could be algebraically tedious. However, it does not stop students from using the GC to check their intermediate working as well as their final answers for z and w.
(b) **(i)** **(ii)**		To show the geometrical relationship of *A, B* and *C*, we should indicate on the diagram that *OC* is the diagonal of the parallelogram *OACB*. Addition of complex numbers is similar to addition of vectors, where we add the real and imaginary parts separately, like we do component wise for vectors.

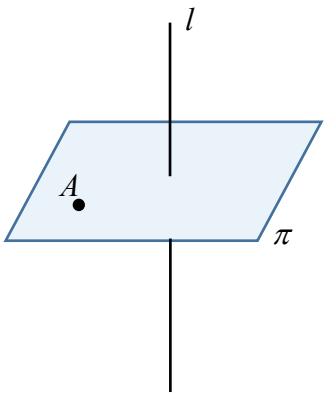
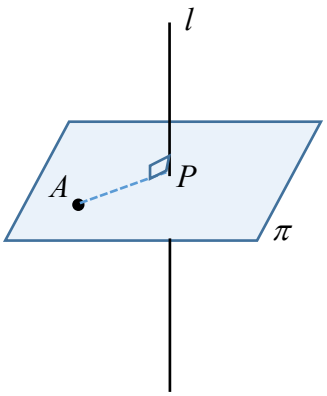
		Effort should be made to ensure OA, OA' are of the same distance from origin, similarly for OB, OB' and OC, OC'.
(b) (ii)	An anticlockwise rotation of $\frac{\pi}{2}$ radians about the origin.	By multiplying i to w, it will rotate OA anti-clockwise by $\frac{\pi}{2}$ radians about the origin.
(b) (iii)	The transformation that maps A to A'' is an anti-clockwise rotation of π radians about the origin. Hence, A'' does not correspond to the complex conjugate of w_1, which would have involved a reflection of point A about the real axis instead.	Alternatively, if complex number v represents point A'', $v = \mathrm{i}(\mathrm{i}w_1) = \mathrm{i}^2 w_1 = -w_1$, which is not equal to $(w_1)^*$.
(b) (iv)	A clockwise rotation of $\frac{\pi}{2}$ radians about the origin. OR An anti-clockwise rotation of $\frac{3\pi}{2}$ radians about the origin.	

No.	Suggested Solution	Remarks for Student
8(a)	$x = a\cos^3 t \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = -3a\sin t\cos^2 t$ $y = a\sin^3 t \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}t} = 3a\cos t\sin^2 t$ $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3a\cos t\sin^2 t}{-3a\sin t\cos^2 t} = -\frac{\sin t}{\cos t} = -\tan t$ At point P with parameter p, (i.e. $t = p$), the equation of the tangent is $y - a\sin^3 p = -\tan p\left(x - a\cos^3 p\right)$ $y = -\tan p\left(x - a\cos^3 p\right) + a\sin^3 p$ $y = -x\tan p + a\sin p\cos^2 p + a\sin^3 p$ $y = -x\tan p + a\sin p\left(\cos^2 p + \sin^2 p\right)$ $y = -x\tan p + a\sin p$	This question contains parametric equations involving trigonometric functions which students have to handle with caution. Note : At point P, $\frac{\mathrm{d}y}{\mathrm{d}x} = -\tan p$

8(b)	At A, $y=0$, $-x\tan p + a\sin p = 0$ $\Rightarrow x = a\cos p$ At B, $x=0$, $y = a\sin p$ $AB = \sqrt{a^2\sin^2 p + a^2\cos^2 p}$ $= a\sqrt{\sin^2 p + \cos^2 p}$ (since $a>0$) $= a$ Since $AB = a$, it depends on a only. (shown)	 Considering ΔAOB, the length of AB can be found easily using Pythagoras' Theorem.
8(c)	$x = a\cos^3 t$ and $y = a\sin^3 t$ $\Rightarrow \cos^2 t = \left(\frac{x}{a}\right)^{\frac{2}{3}}$ and $\sin^2 t = \left(\frac{y}{a}\right)^{\frac{2}{3}}$ Since $\cos^2 t + \sin^2 t = 1$, $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$, $0 \le x \le a$, $0 \le y \le a$ (Cartesian equation of C)	Students need to recall the relevant trigonometric identities to obtain the cartesian equation. Students can derive the range of x and y values from $0 \le t \le \frac{1}{2}\pi$. Eg. $0 \le \cos t \le 1$ $0 \le a\cos^3 t \le a$
8(d)	$x^{\frac{2}{3}} = a^{\frac{2}{3}} - y^{\frac{2}{3}}$ $x^2 = \left(a^{\frac{2}{3}} - y^{\frac{2}{3}}\right)^3$ $= a^2 - 3a^{\frac{4}{3}}y^{\frac{2}{3}} + 3a^{\frac{2}{3}}y^{\frac{4}{3}} - y^2$ Volume of revolution $= \pi\int_0^a x^2 \mathrm{d}y$ $= \pi\int_0^a \left(a^2 - 3a^{\frac{4}{3}}y^{\frac{2}{3}} + 3a^{\frac{2}{3}}y^{\frac{4}{3}} - y^2\right)\mathrm{d}y$ $= \pi\left[a^2 y - \frac{9}{5}a^{\frac{4}{3}}y^{\frac{5}{3}} + \frac{9}{7}a^{\frac{2}{3}}y^{\frac{7}{3}} - \frac{1}{3}y^3\right]_0^a$ $= \pi\left(a^3 - \frac{9}{5}a^3 + \frac{9}{7}a^3 - \frac{1}{3}a^3\right)$ $= \frac{16}{105}a^3\pi$	Though the sketch of the graph is not required as part of the solution, students should sketch the graph using their GC (by setting a = 1) to identify the required bounded region. DO remember to restrict t to $0 \le t \le \frac{\pi}{2}$ in the GC.  Since the cartesian equation for C has been obtained in (c), the required volume can be obtained by integrating in terms of y.

No.	Suggested Solution	Remarks for Student
9(a)	$x^2 + R^2 = (h+R)^2$ $x^2 = (h+R)^2 - R^2$ $= (h+R+R)(h+R-R)$ $= h(h+2R)$ $= 2hR\left(1+\frac{h}{2R}\right)$ $x = (2hR)^{\frac{1}{2}}\left(1+\frac{h}{2R}\right)^{\frac{1}{2}}$	MP is tangential to the earth, thus $\measuredangle MPO = 90°$ i.e. ΔMPO is a right-angle triangle.
9(b)	$\sin\theta = \frac{x}{R+h}$ $= \left(\frac{1}{R+h}\right)(2hR)^{\frac{1}{2}}\left(1+\frac{h}{2R}\right)^{\frac{1}{2}}$ $= \left(\frac{1}{R\left(1+\frac{h}{R}\right)}\right)(2hR)^{\frac{1}{2}}\left(1+\frac{h}{2R}\right)^{\frac{1}{2}}$ $= \left(\frac{1}{1+\frac{h}{R}}\right)\left(2\left(\frac{h}{R}\right)\right)^{\frac{1}{2}}\left(1+\frac{1}{2}\left(\frac{h}{R}\right)\right)^{\frac{1}{2}}$ $\sin\theta = \left(\frac{1}{1+\alpha}\right)(2\alpha)^{\frac{1}{2}}\left(1+\frac{1}{2}\alpha\right)^{\frac{1}{2}}$ $= (2\alpha)^{\frac{1}{2}}(1+\alpha)^{-1}\left(1+\frac{\alpha}{2}\right)^{\frac{1}{2}}$ $= (2\alpha)^{\frac{1}{2}}(1-\alpha+\ldots)\left(1+\frac{1}{2}\left(\frac{\alpha}{2}\right)+\ldots\right)$ (since $\alpha = \frac{h}{R}$ is small as h is small compared to R) $= (2\alpha)^{\frac{1}{2}}\left(1+\frac{\alpha}{4}-\alpha+\ldots\right)$ $\approx (2\alpha)^{\frac{1}{2}}\left(1-\frac{3}{4}\alpha\right)$ (shown)	In this part of the question, students are required to apply binomial series expansion, as an application of "small angles". Since the final answer is in terms of $\alpha = \frac{h}{R}$, students would need to express $\sin\theta$ in terms of α, with a factor of $(2\alpha)^{\frac{1}{2}}$. This will lead the students to $(1+\alpha)^{-1}$ and $\left(1+\frac{\alpha}{2}\right)^{\frac{1}{2}}$ which require binomial series expansion to be applied.

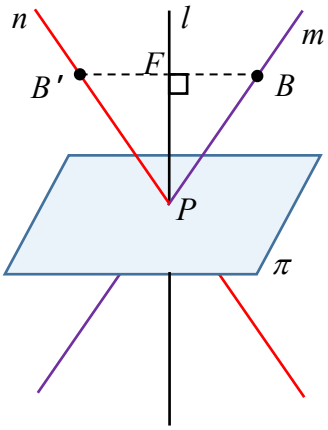
9(c)	Given $\theta = 2°$ and $R = 6375$ km $\sin 2° = \sqrt{2\alpha}\left(1 - \frac{3\alpha}{4}\right)$ Using GC, solving $\sin 2° - \sqrt{2\alpha}\left(1 - \frac{3\alpha}{4}\right) = 0$ yields $\alpha = 6.0954 \times 10^{-4} = 0.000610$ (3 sf) $h = R\alpha$ $= 6.0954 \times 10^{-4}(6375)$ $= 3.89$ km (3 sf)	To sketch the graph with $\sin 2°$, students need to change the mode to "degree" or convert $\sin 2°$ to $\sin \frac{2\pi}{180}$ rads.  After obtaining the graph $y = \sin 2° - \sqrt{2\alpha}\left(1 - \frac{3\alpha}{4}\right)$, find α, the smaller root (out of the 2) that is nearer to 0 (as α is small), by zooming in.

No.	Suggested Solution	Remarks for Student
10(a)	$l: \mathbf{r} = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$ $\pi: \mathbf{r}.\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}.\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ $\mathbf{r}.\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = -9$  The cartesian equation of the plane π is $2x - 3y + z = -9$.	Students need to be able to find the vector equation of line l from its cartesian form. For plane π to be perpendicular to l, its normal vector will be parallel to the direction vector of l.
10(b)	Note that point P is the intersection point between π and l. Since P lies on l, $\overrightarrow{OP} = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$, for some $\lambda \in \mathbb{R}$ Since P also lies in π, $\left[\begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}\right].\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = -9$ $\Rightarrow 4\lambda + (3 + 9\lambda) + (2 + \lambda) = -9$ $\Rightarrow \lambda = -1$ $\overrightarrow{OP} = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$  Coordinates of the point P is $(-2,\ 2,\ 1)$.	Since π is perpendicular to l, the foot of perpendicular of point A to line l lies in π. i.e. AP is the shortest distance from A to l. Students are to leave the final answer for this part in coordinates form.
10(c)	Let $\overrightarrow{OB} = \begin{pmatrix} 4 \\ -5 \\ 10 \end{pmatrix}$ Let F be the foot of perpendicular from B to l. $\overrightarrow{OF} = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$, for some $t \in \mathbb{R}$	It is important for students to be able to visualise the relationship between l, m and n.

$\overrightarrow{BF} = \overrightarrow{OF} - \overrightarrow{OB}$

$$= \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + t\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} - \begin{pmatrix} 4 \\ -5 \\ 10 \end{pmatrix}$$

$$= \begin{pmatrix} 2t-4 \\ -3t+4 \\ t-8 \end{pmatrix}$$



Since BF is perpendicular to l,

$$\begin{pmatrix} 2t-4 \\ -3t+4 \\ t-8 \end{pmatrix} \bullet \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = 0$$

$$-8-12-8+t(4+9+1) = 0$$

$$14t = 28$$

$$t = 2$$

$$\Rightarrow \overrightarrow{OF} = \begin{pmatrix} 4 \\ -7 \\ 4 \end{pmatrix}$$

Let B' be the point of reflection of B about the line l.

By Ratio Theorem,

$$\overrightarrow{OF} = \frac{\overrightarrow{OB} + \overrightarrow{OB'}}{2}$$

$$\overrightarrow{OB'} = 2\overrightarrow{OF} - \overrightarrow{OB} = 2\begin{pmatrix} 4 \\ -7 \\ -4 \end{pmatrix} - \begin{pmatrix} 4 \\ -5 \\ 10 \end{pmatrix} = \begin{pmatrix} 4 \\ -9 \\ -2 \end{pmatrix}$$

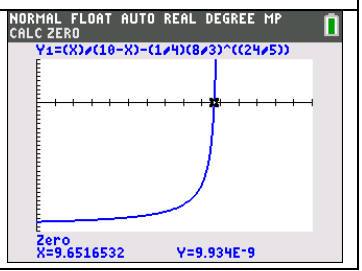
Direction vector for line $n = \overrightarrow{PB'} = \begin{pmatrix} 4 \\ -9 \\ -2 \end{pmatrix} - \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -11 \\ -3 \end{pmatrix}$

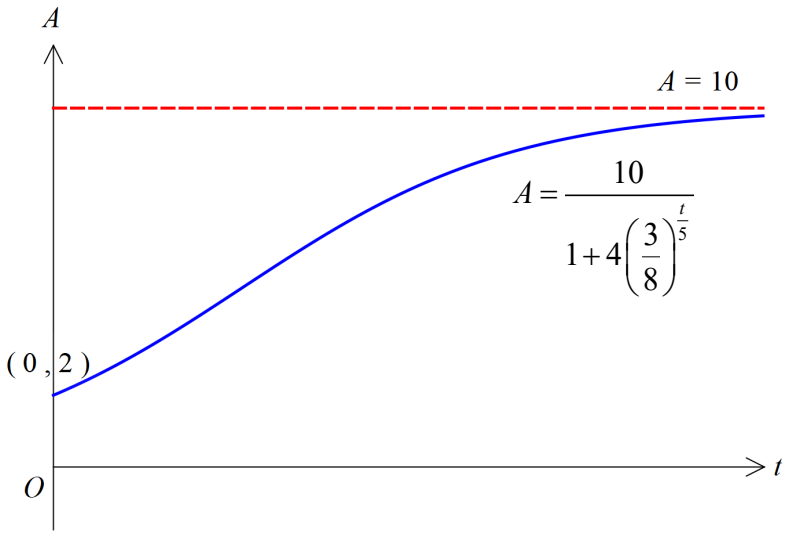
$$n: \ \mathbf{r} = \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} + \mu\begin{pmatrix} 6 \\ -11 \\ -3 \end{pmatrix}, \mu \in \mathbb{R}$$

A cartesian equation for n is $\dfrac{x+2}{6} = \dfrac{y-2}{-11} = \dfrac{z-1}{-3}$.

To find the equation of n, students should aim to find another point other than P that lies on n. One such point will be B', the point of reflection of point B about line l.

After obtaining the vector equation of n, students need to express it in its cartesian form to complete the question.

No.	Suggested Solution	Remarks for Student
11(a)	$\dfrac{\mathrm{d}A}{\mathrm{d}t} = kA(10-A), \quad k>0$ $\displaystyle\int \frac{1}{A(10-A)}\,\mathrm{d}A = k\int 1\,\mathrm{d}t$ $\dfrac{1}{10}\displaystyle\int\left(\frac{1}{A}+\frac{1}{10-A}\right)\mathrm{d}A = k\int 1\,\mathrm{d}t$ Let $\dfrac{1}{A(10-A)} = \dfrac{a}{A}+\dfrac{b}{10-A}$ By cover-up rule, When $A=0$, $a=\dfrac{1}{10}$ When $A=10$, $b=\dfrac{1}{10}$ Since $0<A<10$, $\dfrac{1}{10}\left(\ln A-\ln(10-A)\right) = kt+C$ $\dfrac{1}{10}\ln\left(\dfrac{A}{10-A}\right) = kt+C$ $\ln\left(\dfrac{A}{10-A}\right) = 10kt+10C$ $\dfrac{A}{10-A} = \mathrm{e}^{10kt+10C} = B\mathrm{e}^{10kt}$ where $B=\mathrm{e}^{10C}$ When $t=0$, $A=2 \Rightarrow \dfrac{2}{8} = B\mathrm{e}^{0} \Rightarrow B=\dfrac{1}{4}$ When $t=5$, $A=4 \Rightarrow \dfrac{4}{6} = \dfrac{1}{4}\left(\mathrm{e}^{10k}\right)^5 \Rightarrow \mathrm{e}^{10k} = \left(\dfrac{8}{3}\right)^{\frac{1}{5}}$ $\therefore \dfrac{A}{10-A} = \dfrac{1}{4}\left(\dfrac{8}{3}\right)^{\frac{t}{5}}$ When 80% of the pond is covered in weeds, $A=8$, $\dfrac{8}{10-8} = \dfrac{1}{4}\left(\dfrac{8}{3}\right)^{\frac{t}{5}} \Rightarrow \left(\dfrac{8}{3}\right)^{\frac{t}{5}} = 16$ $t = \dfrac{5\ln 16}{\ln\frac{8}{3}} = 14.13$ (2 d.p)	Students are expected to be able to formulate the DE in accordance to the given question. Using the condition that $0<A<10$ will allow students to simplify their algebraic manipulations (without modulus being involved). We will leave the work of expressing A in terms of t to (iii) of this question. The aim of (i) is to solve for t given A only.
11(b)	After 2 years (i.e. $t=24$), $\dfrac{A}{10-A} = \dfrac{1}{4}\left(\dfrac{8}{3}\right)^{\frac{t}{5}} \Rightarrow \dfrac{A}{10-A} = \dfrac{1}{4}\left(\dfrac{8}{3}\right)^{\frac{24}{5}}$ Using GC, $A=9.65$ (2 d.p)	

11(c)	$\frac{A}{10-A}=\frac{1}{4}\left(\frac{8}{3}\right)^{\frac{t}{5}}$ $\frac{10-A}{A}=4\left(\frac{3}{8}\right)^{\frac{t}{5}}$ $\frac{10}{A}-1=4\left(\frac{3}{8}\right)^{\frac{t}{5}}$ $\frac{10}{A}=1+4\left(\frac{3}{8}\right)^{\frac{t}{5}}$ $A=\frac{10}{1+4\left(\frac{3}{8}\right)^{\frac{t}{5}}}$  	To solve the DE, students are required to express A in terms of t (as mentioned in the question). The solution curve should only show $t>0$, the vertical intercept as well as the long run behaviour (tending towards $A=10$).